

Chapter 6 — Case studies in EDA

Domain narratives show how summaries, plots, and quality checks combine

Learning objectives

- Outline EDA steps for retail segmentation and clinical outcome tables
- Identify exploration priorities in fraud and climate settings

Retail sales analysis

- Mixed currencies
- EDA profiles missingness

Healthcare data analysis

- Clinical records mix labs, history
- Box plots and IQR screens flag vitals

Fraud detection

- Transaction EDA emphasizes extreme amounts, rare positive class counts, and time patterns
- Example 6.7 places outliers in a financial setting
- Open the example 7 module for fraud-oriented summaries

E-commerce segmentation and climate

- Customer segmentation repeats retail motifs, RFM-style features, geography
- Climate monitoring stresses temporal plots, missing sensors

Example 6.5 — Sales and weather hypothesis

- Example 6.5 — hands-on module
- Example 6.5 shows how exploratory correlation can suggest seasonal hypotheses worth later
- Explore the chapter example module
- View files: ``modules/chapter6/example5/``

Takeaways

- Case studies share a pattern: profile, visualize, detect issues, clean, re-explore
- Domain dictates which defects and relationships matter most

Next

- Complete the quiz for this part
- Continue to advanced topics and future directions